

Supplementary Tables for *Effective Archimedean Stark Theorems over Real Quadratic Fields*

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Scope and archive

This supplement contains the record map and the complete Artin-label interval replay referenced by the main paper. It also records queue-level Engine-A statistics that are not used in the uniform theorem.

Archive DOI. <https://doi.org/10.5281/zenodo.21713178>. The main paper PDF and $\text{\LaTeX}$ source, this supplement, and the frozen companion archive are top-level files in the public record.

Supplementary Table S1: certificate record map

result	record
first order six	data/q7-p7-case-v1.json
second order six	data/q14-p7-case-v1.json
RQ-000108	data/rq000108-case-v1.json
RQ-000021	data/rq000021-case-v1.json
ramified-prime-3 control	data/q57-norm27-case-v1.json
RQ-002955	data/rq002955-case-v1.json
order ten	data/q33-p11-order10-case-v1.json
cyclic-quartic character selection	artifacts/engine-c-character-selection-v1.json
$\mathbb{Q}(\sqrt{35})$ CM packet	artifacts/engine-c-w3-tranche-01-verified-v1.json
$e = 6$ powered-unit record	artifacts/engine-c-e6-tranche-01-verified-v1.json
$e = 6$ primitive correction	artifacts/engine-c-e6-primitive-packet-correction-v1.json
$\mathbb{Q}(\sqrt{6})$, $e = 8$ proof route	data/q6-norm8-case-v3.json
Engine-C scope correction	artifacts/engine-c-claim-scope-correction-v1.json
Engine-C Fourier-sign correction	artifacts/engine-c-fourier-convention-correction-v1.json
dual-route packet	data/rq000458-dual-case-v1.json
all archimedean places	artifacts/engine-b-archimedean-place-audit-v1.json
quadratic theorem	data/engine-a-uniform-theorem-v1.json
quadratic Euler degeneracy audit	artifacts/engine-a-euler-degeneracy-v1.json
Shintani source map	artifacts/shintani-1978-source-map-v1.json
index-parity theorem	artifacts/results-paper-index-parity-lemma-v1.json
parity audit	artifacts/results-paper-odd-index-parity-audit-v1.json
literature perimeter	data/literature-perimeter-v1.json

result	record
Roblot sextic-overlap audit	artifacts/roblot-sextic-overlap-audit-v1.json

Supplementary Table S3: relation to Roblot's sextic theorem

Here H/K is the cyclic sextic one-place ray field. The columns record Roblot's hypotheses (A1)–(A3), the certified class number of H , and applicability of [1, Thm. 7.1]. Every class number was computed with `bnfinit(P,1)` and certified by `bnfcertify(bnf)=1`; the local assertion was checked in H/H^+ by exact prime decomposition. Roblot's (A4) is not part of the existence statement in Theorem 7.1; as his footnote to (P2) explains, (A4) is needed instead to deduce (P2) from a conjecturally given Stark unit. It is therefore neither assumed nor silently included in the applicability column below.

case	A1	A2	A3	h_H	Thm. 7.1
RQ-000190	yes	yes	yes	1	applies
RQ-000419	yes	yes	yes	1	applies
RQ-000021	yes	yes	yes	1	applies
RQ-002057	yes	yes	yes	1	no: wild at 3
RQ-002955	yes	yes	yes	1	applies

For RQ-002057 the relative ramification index above 3 is 6. For the other four cases no prime above 3 ramifies in H/K .

Supplementary Table S2: complete Artin-label interval replay

For each cyclic ray group below, r denotes the power of the generator in the pinned PARI ray coordinates. Each rational interval contains exactly one positive root of the absolute packet polynomial.

case	r : isolating interval
RQ-000190	0 : (5.325, 5.326), 1 : (2.713, 2.714), 2 : (2.251, 2.252), 3 : (.187, .188), 4 : (.368, .369), 5 : (.444, .445)
RQ-000419	0 : (20.431, 20.432), 1 : (.189, .190), 2 : (1.090, 1.091), 3 : (.048, .049), 4 : (5.289, 5.290), 5 : (.916, .917)
RQ-000108	0 : (7.892, 7.893), 1 : (2.242, 2.243), 2 : (.126, .127), 3 : (.445, .446)
RQ-000021	0 : (6.109, 6.110), 1 : (7.496, 7.497), 2 : (4.351, 4.352), 3 : (.163, .164), 4 : (.133, .134), 5 : (.229, .230)
RQ-002057	0 : (5.709, 5.710), 1 : (.028, .029), 2 : (.035, .036), 3 : (.175, .176), 4 : (34.732, 34.733), 5 : (27.792, 27.793)
RQ-002955	0 : (18.232, 18.233), 1 : (2.180, 2.181), 2 : (8.620, 8.621), 3 : (.054, .055), 4 : (.458, .459), 5 : (.116, .117)
RQ-001107	0 : (8.645, 8.646), 1 : (.276, .277), 2 : (1.245, 1.246), 3 : (.571, .572), 4 : (.236, .237), 5 : (.115, .116), 6 : (3.611, 3.612), 7 : (.802, .803), 8 : (1.748, 1.749), 9 : (4.234, 4.235)

Engine-A queue statistics

The uniform theorem does not depend on a finite search range. In the frozen verification queue there are 1,560 rows with nonempty differenced character support, comprising 2,232 supported quadratic character occurrences across 912 distinct character fields. Exact imprimitive Euler-factor evaluation finds 672 zero Euler products, affecting 603 rows. In 346 rows every supported derivative vanishes, so the theorem's product is empty and $X_A = 1$.

References

- [1] X.-F. Roblot, *Index formulae for Stark units and their solutions*, Pacific J. Math. **266** (2013), 391–422, <https://doi.org/10.2140/pjm.2013.266.391>.